

Master/
Orchestrator

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Jacobi Co-Simulation: One Macro Step $t\# \rightarrow t\###$

Phase 1: Set All Inputs

All inputs frozen at $t\#$
using outputs from previous step

set $u\#(t\#)$ from $y(t\###)$

inputs set

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inputs set

set $u\#(t\#)$ from $y(t\###)$

inputs set

Phase 2: Parallel Integration

All SUs integrate
independently from $t\#$ to $t\###$

par

[Parallel Execution]

doStep($t\#, H$)

$x\###^1 = \delta\#(t\#, x\##, u\##)$

$y\###^1 = \lambda\#(t\###, x\###^1)$

outputs $y\#(t\###)$

doStep($t\#, H$)

$x\###^1 = \delta\#(t\#, x\##, u\##)$

$y\###^1 = \lambda\#(t\###, x\###^1)$

outputs $y\#(t\###)$

doStep($t\#, H$)

$x\###^1 = \delta\#(t\#, x\##, u\##)$

$y\###^1 = \lambda\#(t\###, x\###^1)$

outputs $y\#(t\###)$

Exchange outputs at $t\###$

New outputs available
for next macro step

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